

MaxGPA — User Documentation

Project: CIS 422/522 Software Methodologies — Project 1

Group: 7

Authors: Keagan Koski, Hiro Nakae, Peyton Phillips, Robert Mimms

Date: Spring 2026 | **Last Modified:** 2026-05-04

Table of Contents

1. [Introduction](#)
 2. [Interface Overview](#)
 3. [Student Tasks](#)
 4. [Admin Tasks](#)
 5. [Troubleshooting](#)
-

Introduction

MaxGPA is a web application designed for University of Oregon students to plan their degrees strategically. By surfacing historical grade distributions—broken down by instructor—the tool allows students to identify sections with the best historical outcomes and build an optimized course plan before registration begins.

The system also provides an **Admin Portal** for staff to maintain the accuracy of grade data and degree requirements.

Interface Overview

The application is organized into three primary pages, accessible via the top navigation bar:

- **Degree Planner:** The main workspace for building a plan.
- **My Degree Plan:** A library of your saved and exported plans.
- **Admin:** The portal for data management (Staff only).

The Degree Planner Layout

Region	Description

Left Sidebar	Filters for Major , Year , Course Search , and Instructor .
Top Panel	The Requirements Checklist . Tracks progress with color-coded badges.
Main Area	Course Cards . Displays grade distribution charts and ranked instructor lists.

Student Tasks

Task 1: Select your major and see which courses are required

1. In the left sidebar, locate the **Major** section.
2. Select your major (e.g., *Computer Science*).
3. The **Requirements Checklist** will appear at the top, showing grey badges for all required courses.

Task 2: Choose which years of grade data to include

1. In the left sidebar, find the **Year** section.
2. Check the boxes for your desired Academic Years (e.g., **AY24**, **AY25**).
3. Course cards will update to reflect data only from those periods.

Note: At least one year must be selected for data to appear. AY26 refers to Fall 2025 – Summer 2026.

Task 3: Find the best instructor for a required course

1. Find the specific **Course Card** in the main area (e.g., CS 313).
2. Click the card to expand it.
3. Review the **Ranked Instructor List**. The instructor at the top has the highest historical average GPA.

Task 4: Add a course to your plan

1. Expand a course card and find your preferred instructor.
2. Click the **+ Add** button.

3. The course badge in the Checklist will turn **green**, and the course will appear in "**Your Plan**" on the right.

Task 5: Generate an optimal plan automatically

1. Ensure a **Major** and **Year** are selected.
2. Click the purple **Generate Plan** button (sparkle icon) in the Requirements Checklist.
3. The system will automatically select the highest-GPA instructor for all remaining requirements.

Task 6: Save your completed plan

1. Add courses until the progress indicator shows all sections are complete.
2. Click the green **Save Plan** button.
3. Enter a name for your plan in the dialog box and click **Save**.

Task 7: View and export a saved plan to PDF

1. Navigate to **My Degree Plan** in the top bar.
2. Locate your plan card and click **Export PDF**.
3. The file will download automatically.

Admin Tasks

Task 8: Upload new grade-history data

1. Go to the **Admin** page > **Import Grade Data**.
2. Drag and drop your grade-history CSV.
3. Verify the data preview (Terms, Subjects, and Grade Counts).
4. Click **Confirm & Import**.

Task 9: Import a degree plan for a major

1. Go to **Admin > Degree Requirements**.
2. Select the tab for the specific major.
3. Upload the structured degree-plan CSV.
4. Click **Confirm & Import**.

Warning: This action replaces all existing requirements for that major. This cannot be undone. (developer can run: `python db_init.py`)

Troubleshooting

- **No courses appear:** Ensure at least one **Year** is checked in the sidebar.
- **Save Plan button is greyed out:** You have not fulfilled all requirements. Check for grey badges in the Requirements Checklist.
- **PDF export is blank:** Use **Google Chrome** for the best rendering of canvas-based charts.
- **App won't load:** Verify that the Frontend (Port 3000) and Backend (Port 8000) are both active.

AI Disclosure

The development of MaxGPA involved the use of AI language models (including Anthropic Claude) for requirements analysis, implementation, and documentation. All outputs were reviewed and integrated by the human authors listed above.